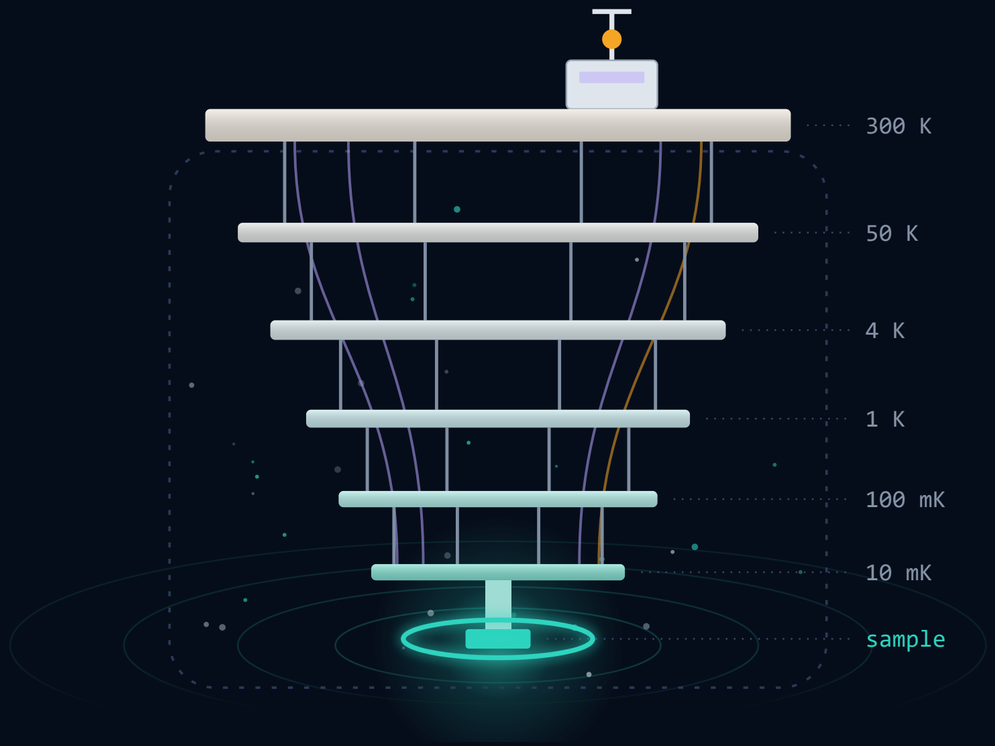
VIBECHECK · THE USABLE FORMAL METHODS
HACKATHON · SAN FRANCISCO · NOV 2026

CryoLean

One experiment. One machine.
One honest answer.

Requirements met **Need more information** **Does not meet**



Hanah Rahman · Nicholas Brandenburg

hrahman12.github.io/CryoLean · github.com/hrahman12/CryoLean · generic cryostat drawn in code, no real model

/ WHAT IT DOES · THREE REAL CHECKS

Three experiments. Three honest answers.

AEROSPACE

Mars-rover sensors

−135 to +70 °C cycles,
+125 °C bake

on a Tinius Olsen chamber

Requirements met

Chamber documents **−150 to +600 °C**.

QUANTUM

300 mm silicon wafers

at ≤ 1.6 K

on a Bluefors XLDHehp

Need more information

Wafer size unconfirmed (**1 of 3** needs). Lean waits.

BIOLOGY

Protein crystal

flash-cooled in
liquid nitrogen

on a Lake Shore CCS-800

Does not meet requirements

No cryogen documented.
A CryoMed TSCM48XV passes.

Verdicts as the website shows them: site/data.json; data/experiments/v4_aero_mars_prt_thermal_cycling.json. The quantum demo leaves wafer size unconfirmed on purpose; confirmed, it passes. Lean checks the plan against sourced specs, not the machine itself.

• LIVE · REAL LEAN VERDICTS

Live: you pick the experiment.

1 **Choose an experiment**
20 industries, 3 experiments each

2 **Choose a cryostat**
220 configurations

3 **Check the setup**
Lean shows the first stage that fails

4 **Ask for a fix**
you only see fixes Lean passes

5 **Break it**
untick a need: **Need more information**



hrahman12.github.io/CryoLean · every answer on the page is one of 138,564 precomputed Lean outcomes (13,200 featured pairs) · Demos tab shown